Fall 2025

Software Requirements Specification

FlipValue

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Github:
https://github.com/lynrasco/PRJ566NBB_2261_Group05

Executive Summary

Background

Description

Company Value Add

End-User Value Add

Scope

What is Included

What is Not Included

Justification

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Section 1

1.1 Document Authors

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1.2 Document Revision History

WEEK	DATE	Revisions
1	January 19, 2026	• Created initial Problem Statement • Defined initial Product Vision • Created public GitHub repo and added members
2	January 22 - 25, 2026	• Added and modified Section 1 • Modified Problem Statement, Product Vision, and added Product Background
3	January 26-31, 2026	• Added stakeholders and users, scope, risks, and operating environment sections • Added probability matrix for risks
4	February 2, 2026	• Created interview questions for users and partner companies
5	February 5-8, 2026	• Added answers to interview questions under functional requirements • Fixed formatting for section 2.1.1 and added sources
6		•
7		•
8		•
9		•
11		•
12		•
13		•
14		•

1.3 Document Purpose

This document serves as a blueprint and framework for FlipValue. All stakeholders will use this as a reference to fully understand how the application works in its intended environment and what it will need for it to work. It should include the following:

- A description of the app's functionality and features
- Any limitations in a specific environment or condition, preferably should satisfy the audience's needs
- Labels any technical and operational requirements such as in this case, the use of AI is required
- Designs to visualize how the application would appear to the viewer through UI/UX designs

1.4 Audience

Resellers

1.5 Group Agreement

Team: #5

Project Title: FlipValue

Project Time Frame:

The entire project should span around two semesters, so roughly January 5, 2026 – April 17, 2026 for the design and planning, and May 4, 2026 – August 14, 2026, for the execution and creation of the actual application.

Team Members:

Joseph Fuh Che,

Aliyah Ighodaro,

Lyndsey Rasco,

Abigael Julao,

Sezim Duishenbieva

Team Leadership:

Joseph Fuh Che

Team Functions/Roles:

We will share information through: Microsoft Teams, Microsoft Word, and in-person meetings.

Programmers & Testers: Joseph Fuh Che, Aliyah Ighodaro, Sezim Duishenbieva

Designers: Abigael Julao, Lyndsey Rasco

Team Meetings:

Utilizing our weekly Thursday presentations as well as Monday lectures:

Monday: 13:30 – 15:15

Thursday: 11:40 – 13:25

We may also meet via Teams at any other time, if necessary.

Team Problems:

Miscommunication: Team must be present during meetings as well as communicating effectively via Teams. If communication between members is unclear and disorganized, it leads to misunderstanding and delayed progress within the group, so report to the team leader if a member (s) struggles to communicate.

Unequal Workloads: An uneven distribution of work among team members is usually caused by unclear roles and responsibilities, an over-reliance on those who are actively contributing, and inadequate leadership. To solve this, we should be able to lay out clear roles and responsibilities prior to doing any of the actual work.

Team Commitment:

Each team member is committed to contributing their parts and findings to the project and participating in all team meetings. If a member is struggling or if any problems occur, it should be communicated to the team, specifically the team leader and should be resolved immediately.

The undersigned members agree to work together on the project until the end of the PRJ666 next Semester. They recognize that as a team and individually they are equally responsible for the quality of all deliverables.

Name	Date	Signature
Abigael Julao	Jan 19, 2026	AMJ
Lyndsey Rasco	Jan 19, 2026	LDTR
Sezim Duishenbieva	Jan 19, 2026	SD
Aliyah Ighodaro	Jan 19, 2026	AI
Joseph Fuh Che	Jan 22, 2026	JFC

Section 2

2.1 Project Proposal

2.1.1 Project Background:

Features	Poshmark	Depop	Karrot	eBay	FlipValue
Item Listing & Selling	✓	✓	✓	✓	
Market-Based Price Suggestions					✓
Real-time Price Trends					✓
AI Price Prediction					✓
Condition-Based Pricing					✓
Cross-Platform Comparison					✓
Historical Sales Analysis				Limited	

Poshmark [1]

Poshmark is a commerce platform focused on fashion resale, allowing users to list, buy, and sell second-hand clothing and accessories. While the platform provides visibility into similar listings, pricing decisions are left completely to the seller. Users must manually compare listings, which may vary in condition, age, and demand. Poshmark does not provide real-time market trends or inaccurate pricing decisions.

Depop [2]

Depop targets a younger demographic and emphasizes heavily in trend-driven sales. The platform provides basic price suggestions based on item category, brand, and condition; however, these recommendations are limited in scope and do not analyze real-time market demand or historical sale trends. Depop does not offer pricing analytics, trends visualization, or predictive pricing tools, requiring sellers to rely on subjective judgement and manual comparisons when setting prices.

Karrot [3]

Karrot focuses on local resale and emphasizes quick, community-based transactions. The platform provides a general pricing guide through educational content, such as tips on negotiation strategies, perceived value, and effecive item presentation. However, this guidance is not data-driven, item specific, does not have historical sales analysis, market demand, or pricing suggestions. As a result, sellers must rely on personal judgement and compare similar listing manually.

eBay [4]

eBay focuses on global resale, offering both fixed pricing and auctions. The platform allows sellers to view completed and sold listings, offering limited insight into past sale prices for similar items. While this feature may help with basic price estimation, the data is not in depth and does not consider sale trends, item condition, age, or brand. Advanced pricing analytics are available through their in-app tools, however they are locked behind a paywall thus the standard user cannot experience pricing support. As a result, seller rely on manual analysis, which may lead to inconsistent or inaccurate pricing.

While existing platforms provide the infrastructure for buying and selling second-hand goods, they offer limited support for pricing decisions. Sellers must manually assess the market value of their items across multiple platforms.

FlipValue proposes a supporting mobile application designed to address this gap by leveraging AI and real-time market data. Rather than acting as a resale marketplace, FlipValue aims to analyze comparable listings, historical sales data, and demand trends across partner platform to generate accurate and competitive price suggestions. This enables seller to make informed pricing decision efficiently, reducing uncertainty and improving overall sales.

2.1.2 Problem Statement

The problem of	fixing the right prices for used products.
affects	Resellers who struggle to price their products due to lack of real-time market data
the impact of which is	Overpricing which can discourage a potential buyer or under-pricing which can lead to loss of revenue.
a successful solution would be	a mobile application infused with AI to help these resellers by suggesting prices curated from partner websites which could easily lead to a sale.

2.1.3 Product Vision

For	Resellers

Who	want to sell items with a fair price
The (product name)	FlipValue
That	helps resellers figure out the reasonable prices for items
Unlike	Poshmark, Depop, and Karrot who only provide the platform through which sales could be made
Our product	Uses real-time market data to suggest fair prices for those items, increasing business operations.

2.2 Stakeholders and Users

Internal:

- Abigael – Designer
- Joseph- Team leader, Programmer/Tester
- Lyndsey- Designer
- Sezim – Programmer/Tester
- Aliyah – Programmer/Tester

External:

- Resellers – Individuals who sell items but struggle with accurate pricing
- Partner Companies – External stakeholders that provide the real resale data used by the AI to suggest fair prices
- AI/ML Service Providers - Third-party APIs and services used for image recognition and pricing analytics
- Shoppers/Buyers - Indirect beneficiaries

Stakeholder Name	Project Role	Internal/External	Engagement Level	Level of Interest	Level of Power
Abigael	Designer	Internal	Leading	High	High
Joseph	Team leader, Programmer/Tester	Internal	Leading	High	High
Lyndsey	Designer	Internal	Leading	High	High
Sezim	Programmer/Tester	Internal	Leading	High	High
Aliyah	Programmer/Tester	Internal	Leading	High	High
Resellers	Individuals who sell items but struggle with accurate pricing	External	Supportive	High	High

Partner Companies	provide the real resale data used by AI to suggest fair prices	External	Neutral	Low	High
AI/ML Service Providers	Third-party APIs/ services used for image recognition and pricing analytics	External	Supportive	Medium	Medium
Shoppers/Buyers	Indirect beneficiaries	External	Unaware	Low	Low

2.3 Project Scope

The scope of FlipValue is to design and propose a mobile application that assists reseller in determining fair and competitive resale prices for second-hand items. The objective of the system is to reduce the time and uncertainty involved in manual market research by utilizing an AI-assisted analysis and comparable real-time resale data. The application focuses on usability, accuracy, and efficiency to make informed pricing decisions and improve profitability.

In Scope:

- Mobile application accessible on iOS and Android
- User account registration and profile management
- Image-based item recognition using AI-assisted analysis
- Price recommendation based on:
    - Similar item listings on resale platforms (Poshmark, Karrot, Depop)
    - Market trends
    - Item attributes (brand, condition, category)
- Display suggested price ranges and resale insights
- Ability so save, edit, and manage listings
- Market comparison summaries from selected resale platforms
- Basic analytics for tracking pricing history and sale performance
- Cloud-hosted backend for data storage

Out of Scope:

- Direct buying or selling of items within the application
- Payment processing or financial transactions

- Messaging between buyers and sellers
- Full integration with all third-party resale platforms

2.4 System Risks

Market Risks:

1. Competition from other companies resulting in the project's irrelevance
2. Possible lack of customer demand for the product

Financial Risks:

3. Increases in material costs due to inflation
4. Poor budget planning

Technical Risks:

5. Performance failures due to bugs in the software
6. Data/Security breaches, especially due to the use of AI in our application
7. AI isn't always reliable, and may be biased towards certain companies and their items
8. Since we rely on third-party APIs, they may change and synchronization isn't always possible

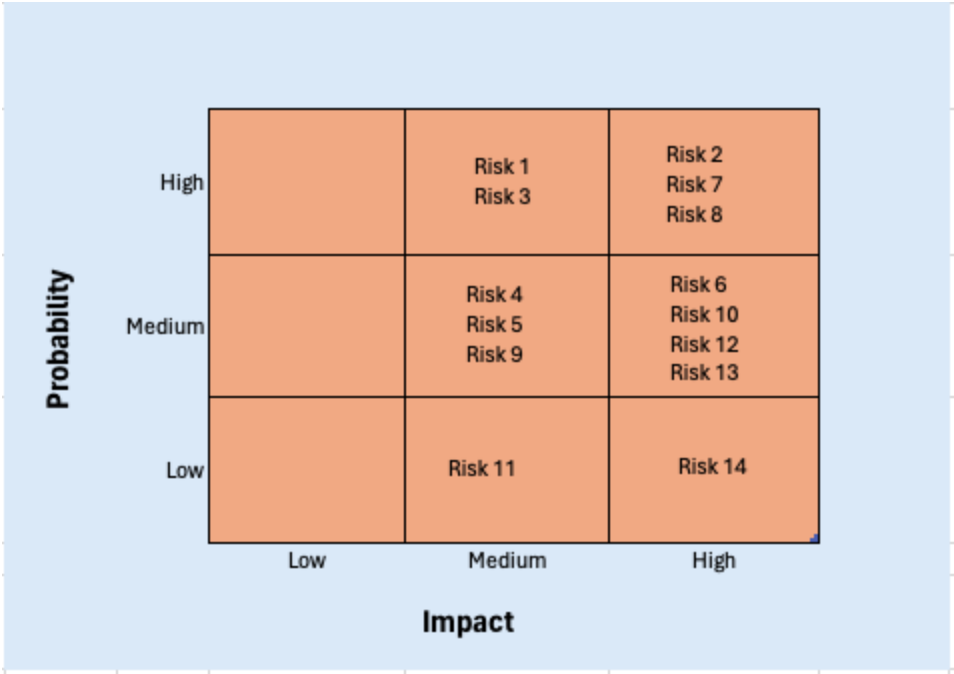
People Risks:

9. Poor communication between stakeholders
10. Incompetent team members result in delayed and poor progress
11. Conflict within team members as well as other stakeholders
12. Team members going on break or getting a Co-op placement next semester

Structure/Process Risks:

13. Inadequate testing procedures for quality control
14. Lack of structure within the organization, such as not establishing clear roles and responsibilities

Probability/Impact Matrix:



2.5 Operating Environment

The FlipValue application is intended to run in a mobile and cloud-based environment. Since the primary users are resellers who mainly use smartphones, the system is designed to be accessible on common mobile devices and operating systems.

Hardware / Devices Supported

- Smartphones and tablets with camera functionality
- Devices capable of installing modern mobile applications

Operating Systems

- Android (version 10 or later)
- iOS (version 14 or later)

Software Dependencies

- Mobile application framework for cross-platform development
- Cloud-based backend services for data storage and processing
- Third-party AI/ML services for image recognition and pricing analysis
- Secure user authentication services

Internet Connection

- An internet connection is required to use core features such as image uploads, AI price suggestions, and real-time market data access
- Limited offline access may be available for viewing saved items only

2.6 Functional Requirements

User Interviews:

- Name: Jane Doe
- Age: 28
- Status: Resells clothing and accessories on Depop and Poshmark
- Financial Situation: Moderate, relies on reselling as supplemental income

As a reseller, how many times have you inaccurately priced clothing and accessories? What were those experiences like and how can FlipValue prevent that?

"Plenty of times. Sometimes, I forget to do my research and underestimate my item's conditions. I also get too sentimental with some of my things that I often set them higher, even lower prices. On one occasion, I priced a pair of jeans at a higher price than it should, thinking that was a fair decision, but later discovered that they had some stains on them. I hope that FlipValue can help prevent that by helping me compare prices of similar items from other different brands, as well as giving me a better idea on how to price my things."

We are using AI to help resellers, such as yourself, to make better decisions when selling items. What would make you trust/not trust their suggestions?

"If the AI model were to give me their reasoning behind their suggestions, such as trends, then yeah, I would trust it, because it shows some transparency rather than giving me the suggestion, I want to know the 'why'. If AI wasn't so transparent, I would think twice before taking their suggestion to account. I would still like to be able to choose for myself, so I am hoping that FlipValue does not force us to follow the AI's suggestions blindly and we still have the choice if we want to accept or even adjust them. As a user, I would like my confidentiality honored, so if the AI starts asking questions such as personal info, it will feel like an invasion of privacy and would make me feel less confident about accepting their suggestions."

What features would you like to see on the dashboard? Are there any deal breakers or must-haves you would like us to know?

"Honestly, I want the dashboard to be minimalistic. I don't want the dashboard to be cluttered with information, specifically unnecessary information; it would be difficult to navigate through the app. Visuals are a huge deal breaker for me. I wouldn't mind some colour, such as a muted and consistent colour palette and font designs, not super contrasted. Another feature I would love to see are filters; it would make my life so much easier without me having to manually type out what I want to see and helps make me better decisions."

Partner Companies Interviews:

- Name: John Doe
- Status: Shares information on clothing and accessories
- Occupation: Data Partnership Manager
- Age: 51
- Financial Situation: Stable, focused on sustainable revenue growth

What data/information should our AI model track? Are there certain things that the AI model should or shouldn't access?

"Our company can share basic information like item type, brand, condition, selling price and how long it took to sell, and general location. This helps your AI understand market trends and give better price suggestions. However, we will not share personal user information, private messages, or payment details. User privacy must always be protected."

What concerns might prevent your organization from entering this partnership?

"Our biggest concerns are data safety and privacy. We want to be sure our information is not misused or shared without our permission. We are also worried about how much time and effort it takes to connect our systems. If the integration process is too difficult or risky, we may not want to participate. Clear rules and agreements would help us feel more comfortable."

Would you prefer your platform to be visible to users or used only in background analysis?

"I would prefer our platform to be visible to users as one of the sources used for price comparisons. This helps build trust because users can see where the information is coming from. It also gives our company more recognition and shows that we are a part of a reliable system. At the same time, I am okay with our data being used in the background if it helps improve accuracy and protects privacy. As long as our data is being used responsibly, kept secure, and properly credited, both options can work well for us"

2.6.1 Core Functional Requirements

This section outlines the main functions that the FlipValue system must support to help resellers accurately price their items. These requirements are based on user and partner interviews as well as the defined project scope.

User Account Management

- The system shall allow users to create an account using an email and password.
- The system shall allow users to log in and log out of their account.
- The system shall allow users to view and update basic profile information.
- The system shall retain user data and preferences across sessions.

Item Entry and Management

- The system shall allow users to manually enter item details such as category, brand, condition, and description.
- The system shall allow users to upload images of items using their mobile device.
- The system shall allow users to save item listings for future reference.
- The system shall allow users to edit or delete saved items.

AI-Assisted Item Analysis

- The system shall analyze uploaded images to identify item characteristics such as brand, style, and material.
- The system shall use both image analysis and user-provided information to improve pricing accuracy.
- The system shall ensure that AI analysis only uses data necessary for pricing purposes.

Price Recommendation

- The system shall collect comparable resale data from partner platforms.
- The system shall generate a suggested resale price range based on market data and item attributes.
- The system shall display basic factors influencing the suggested price, such as item condition or demand.
- The system shall allow users to adjust or ignore the suggested price if desired.

Market Comparison and Insights

- The system shall display comparable prices from selected resale platforms.
- The system shall provide simple market trend insights for similar items.
- The system shall indicate whether an item is priced below, above, or within the average market range.

User Dashboard

- The system shall provide a dashboard displaying all saved items and their suggested prices.
- The system shall allow users to filter and sort items by category or price.
- The system shall present information in a clean and easy-to-understand layout.

Data Storage and Retrieval

- The system shall securely store user accounts, item information, images, and pricing history.
- The system shall retrieve stored data when the user logs into the application.
- The system shall keep stored data consistent and up to date.

Partner Data Integration

- The system shall use resale data provided by approved partner companies for pricing analysis.
- The system shall ensure that partner data is used only for comparison and pricing purposes.

2.7 Nonfunctional Requirements

1. Usability:
   - Simple and minimalistic UI
   - User Friendly

2. Reliability:
   - The application works correctly over time with low failure rates.
   - The application handles errors gracefully
3. Security:
   - Authentication and authorization.
   - Protection against cyber-attacks.
   - Data encryption.

4. Maintainability:
   - Easy to fix bugs.
   - Easy to update and improve.
   - Well documented code following the approved layers of programming to ease maintenance.

5. Portability:
   - Runs on many platforms
   - Minimal changes require during environment change.

6. **Compatibility**
   - Works with other systems, APIs, browsers, AI, or devices.
   - Backward compatibility.

7. Efficiency:
   - Optimal use of CPU, memory, battery and network.
   - Data is gathered, processed and displayed to the users withing seconds

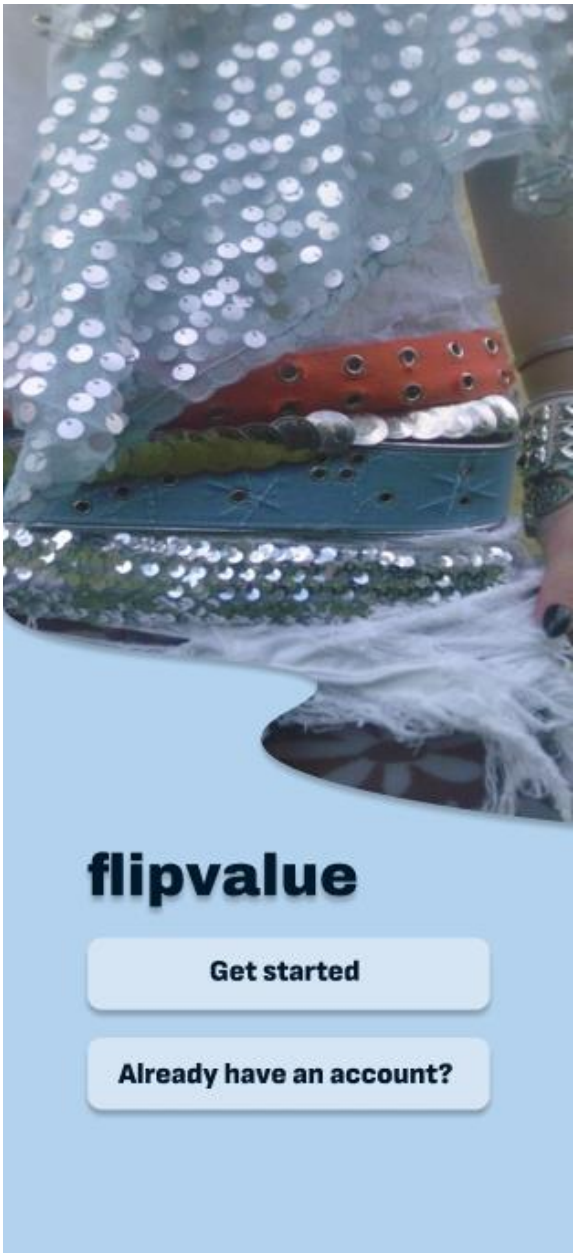
8. Legal Compliance:
   - Meets laws, standards and regulations.

9. Availability:
   - Accessible at all times.

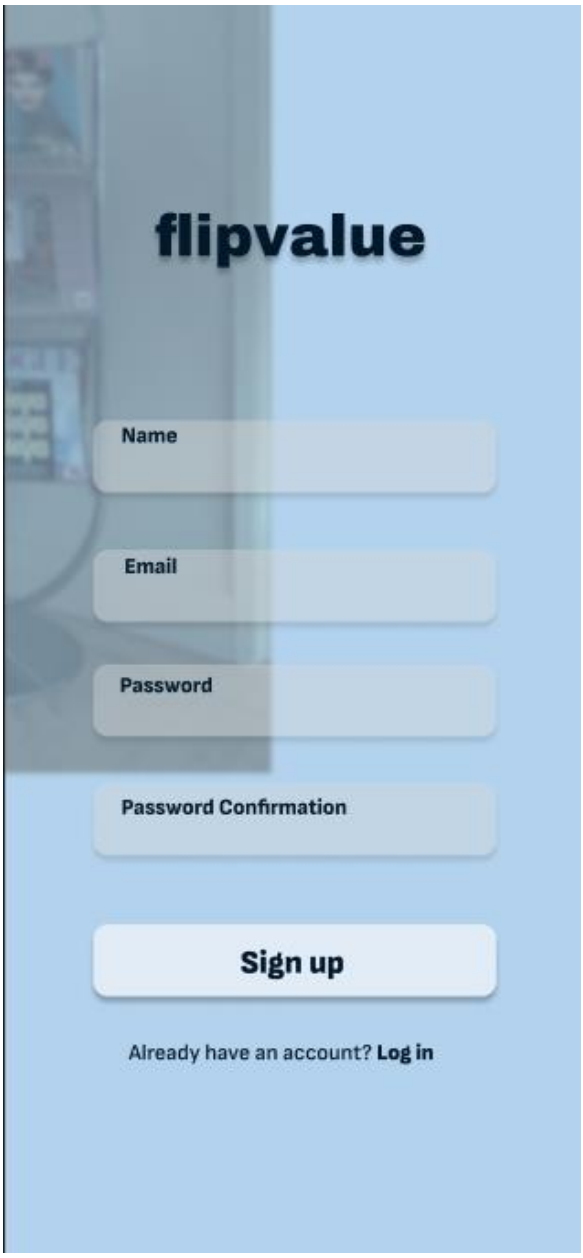
2.8 UI/UX Interface Mock-ups

Figma Project: [Figma Link FlipValue]

Welcome Page:



Sign Up:


flipvalue
Name
Email
Password
Password Confirmation
Sign up
Already have an account? Log in

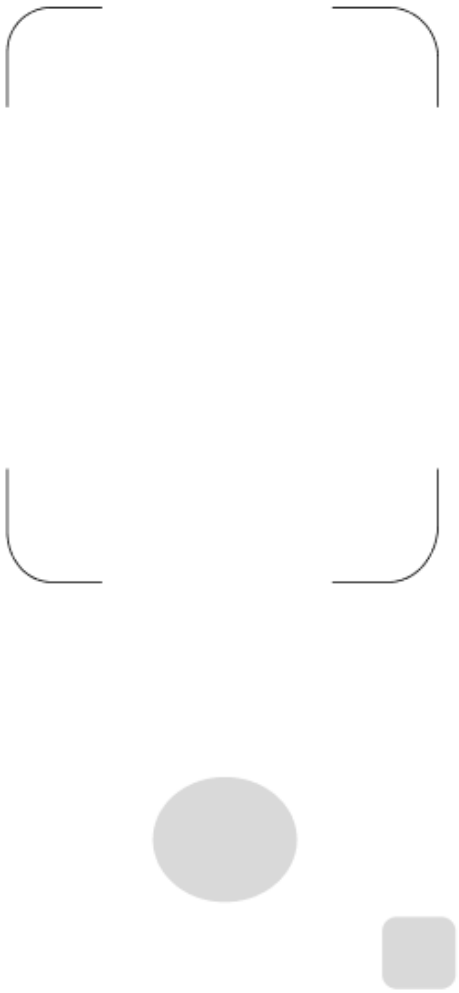
Log In:

flipvalue
Email
Password
Log in
Don't have an account? Sign up

Homepage:

Camera:

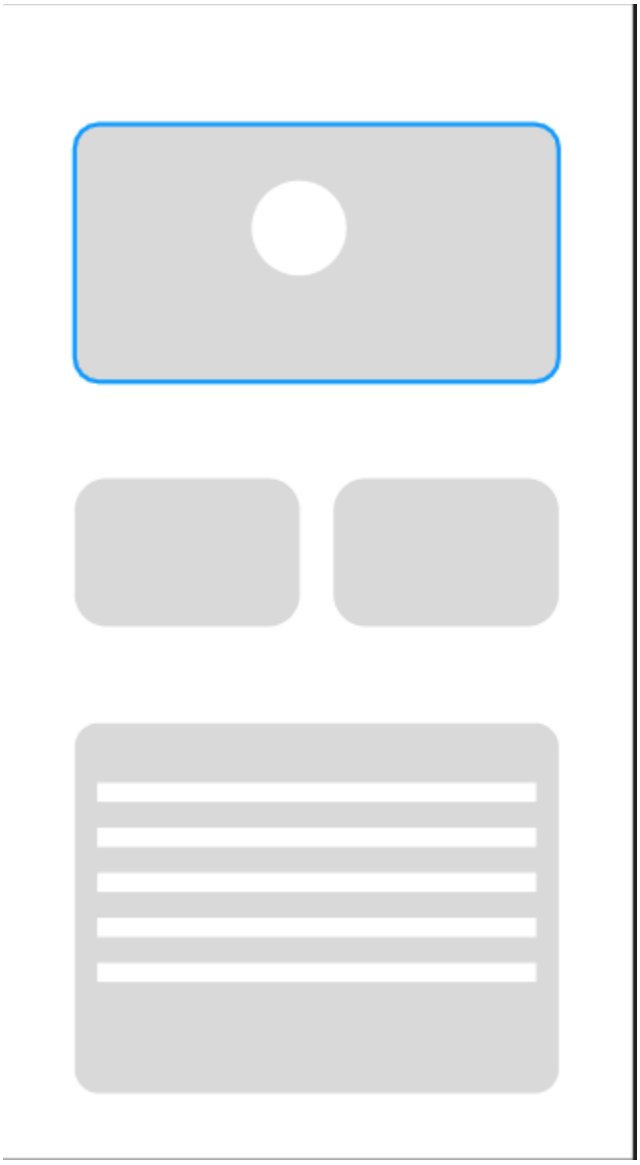
X



Item details:

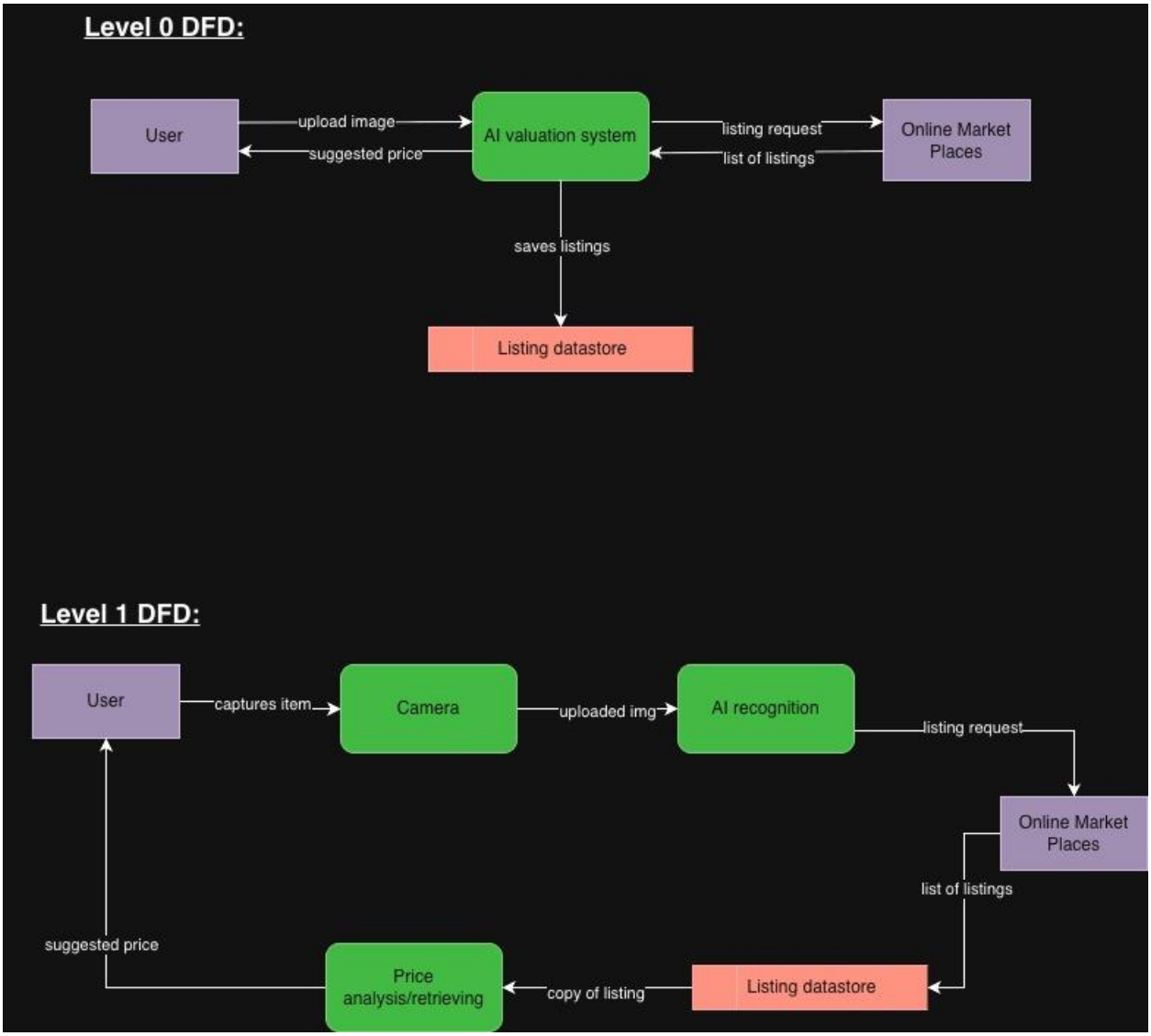
Listings from Partners:

Settings:



Section 3

3.1 Data Flow Diagrams



3.2 User Stories and related Use Case Scenarios

Use Case #	Use Case Name	Actor(s)	Trigger	Outcome	Priority
UC-1	Login/Authenticate User	User	The user wants to log in to their account	The user has successfully logged in to FlipValue	High

UC-2	Enter Item Details	User	The user requests to enter the details of their item	Item details have been entered and filled in.	High
UC-3	Request Market Comparison Insight	User	The user asks the AI model for a market comparison insight for their item	The AI successfully displays pricing trends, average prices, and other similar items	High
UC-4	Generate/Price Suggestion	User, AI/ML Service Provider	The user wants a pricing suggestion when selling their item.	The AI model successfully gives the user a suggested price for the item	High
UC-5	View Data Analytics	Partner Company	The partner company wants to see how the AI model is using their info	The app shows AI usage trends, and other analytics on the dashboard	High

3.2.1 Login/Authenticate User

User Story: As a user, I want to log in to my FlipValue account so that I can start listing my items for sale.

3.2.2 Enter Item Details

User Story: As a user, I want to enter the details of my item so that the AI model can provide pricing suggestions.

3.2.3 View Market Comparison Insight

User Story: As a user, I want to request a market comparison analysis generated from the AI so that I can evaluate market trends and see how the price of my item compares to other similar items.

3.2.4 Generate/Price Suggestion

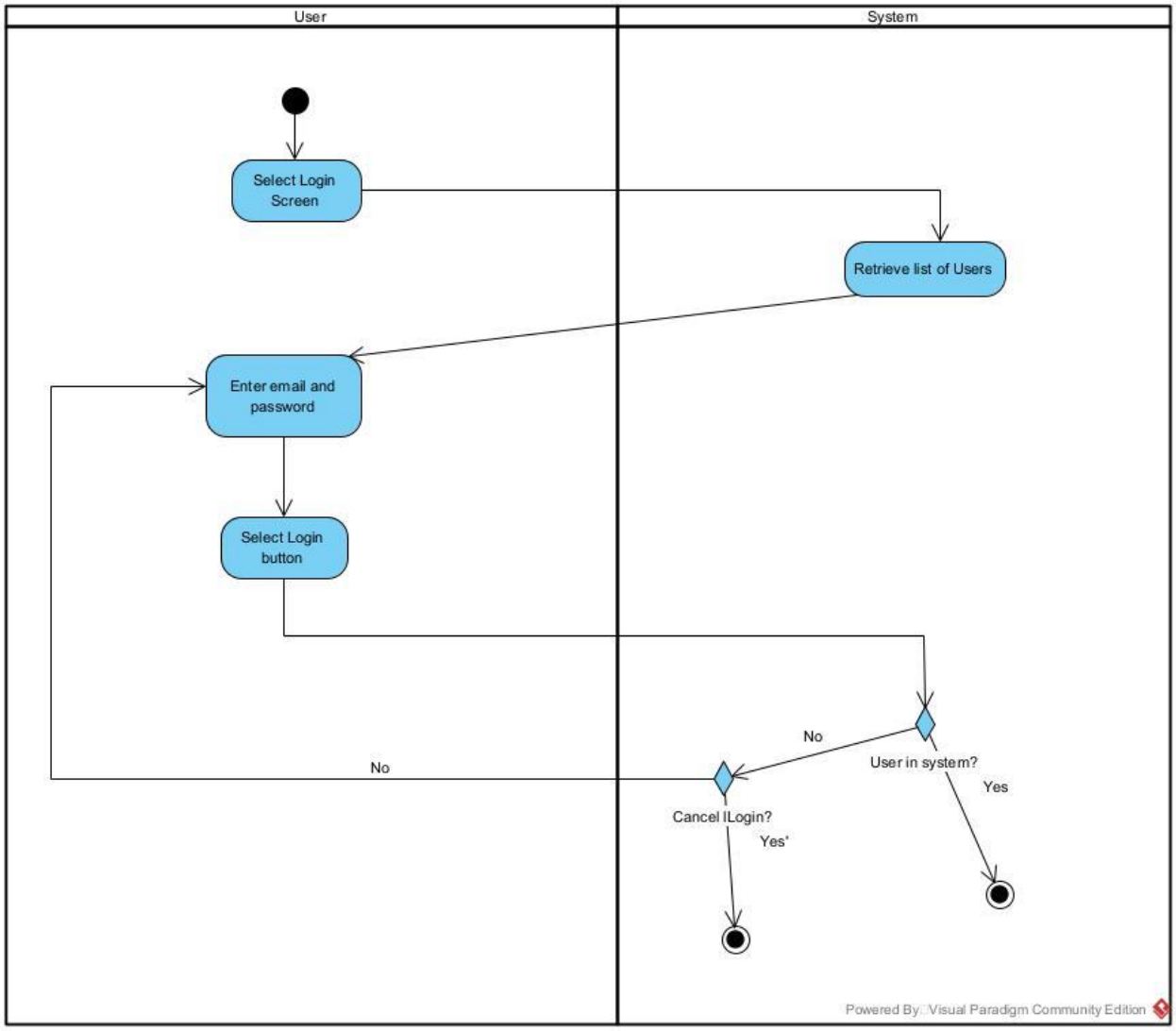
User Story: As a user, I want to request the AI to generate a pricing suggestion for me so that I can make better listing decisions.

3.2.5 View Data Analytics

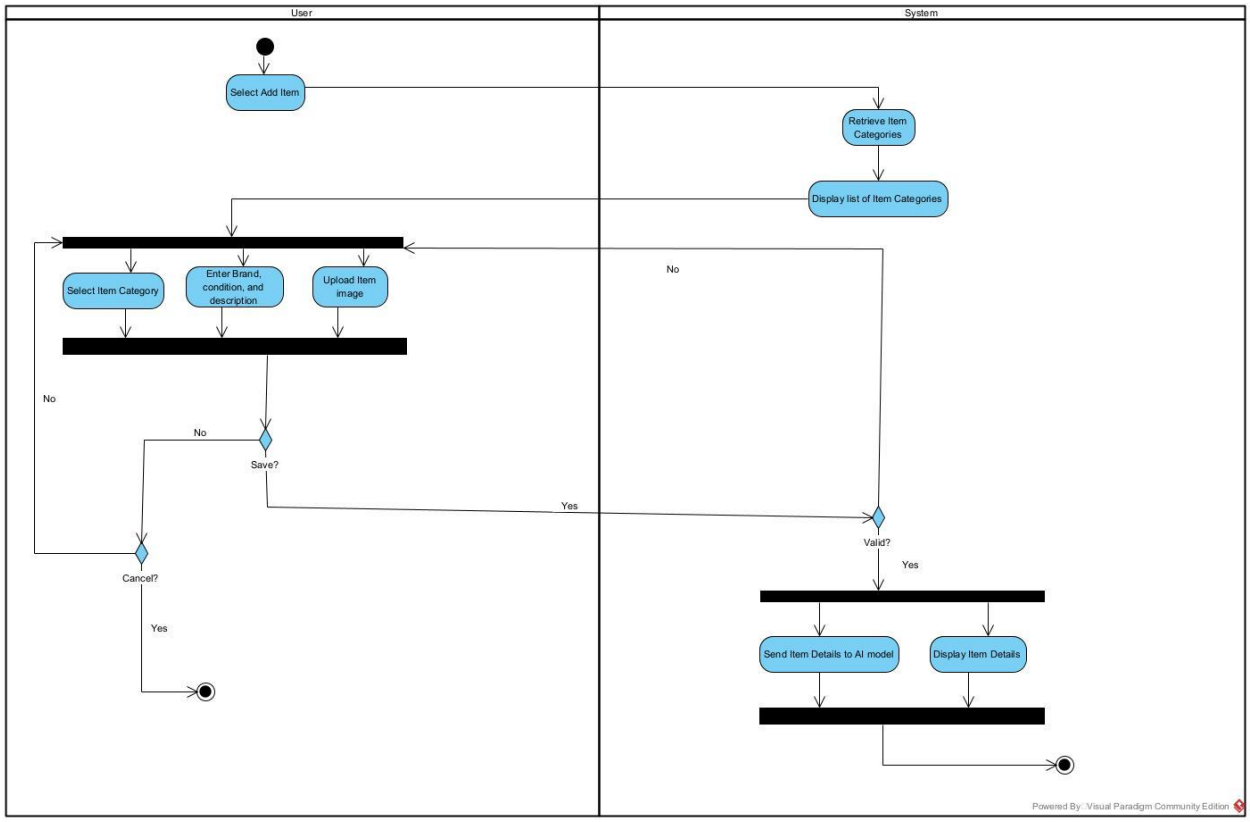
User Story: As the manager of a partner company, I want to view the data analytics so that I can see how the AI is utilizing our data in pricing suggestions and ensure transparency.

3.3 Activity Diagrams

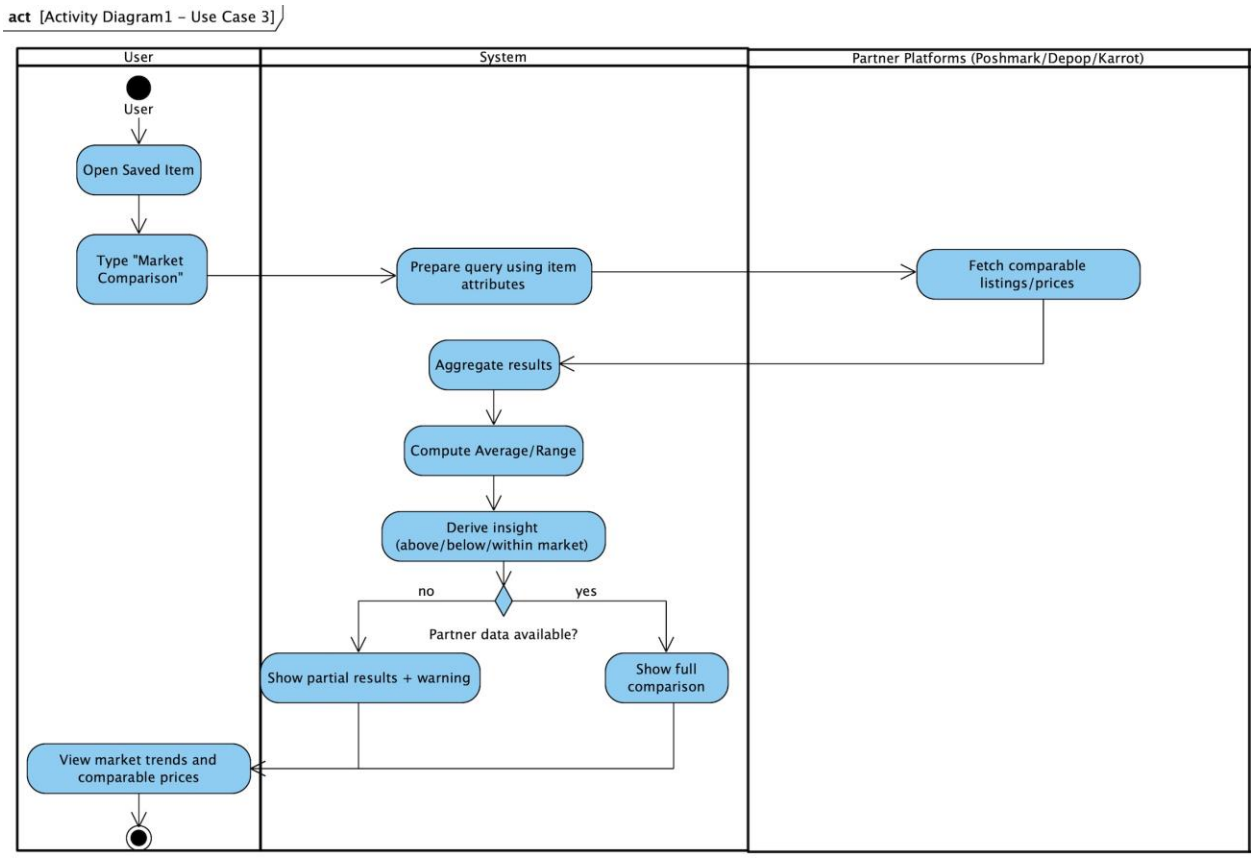
AD1:



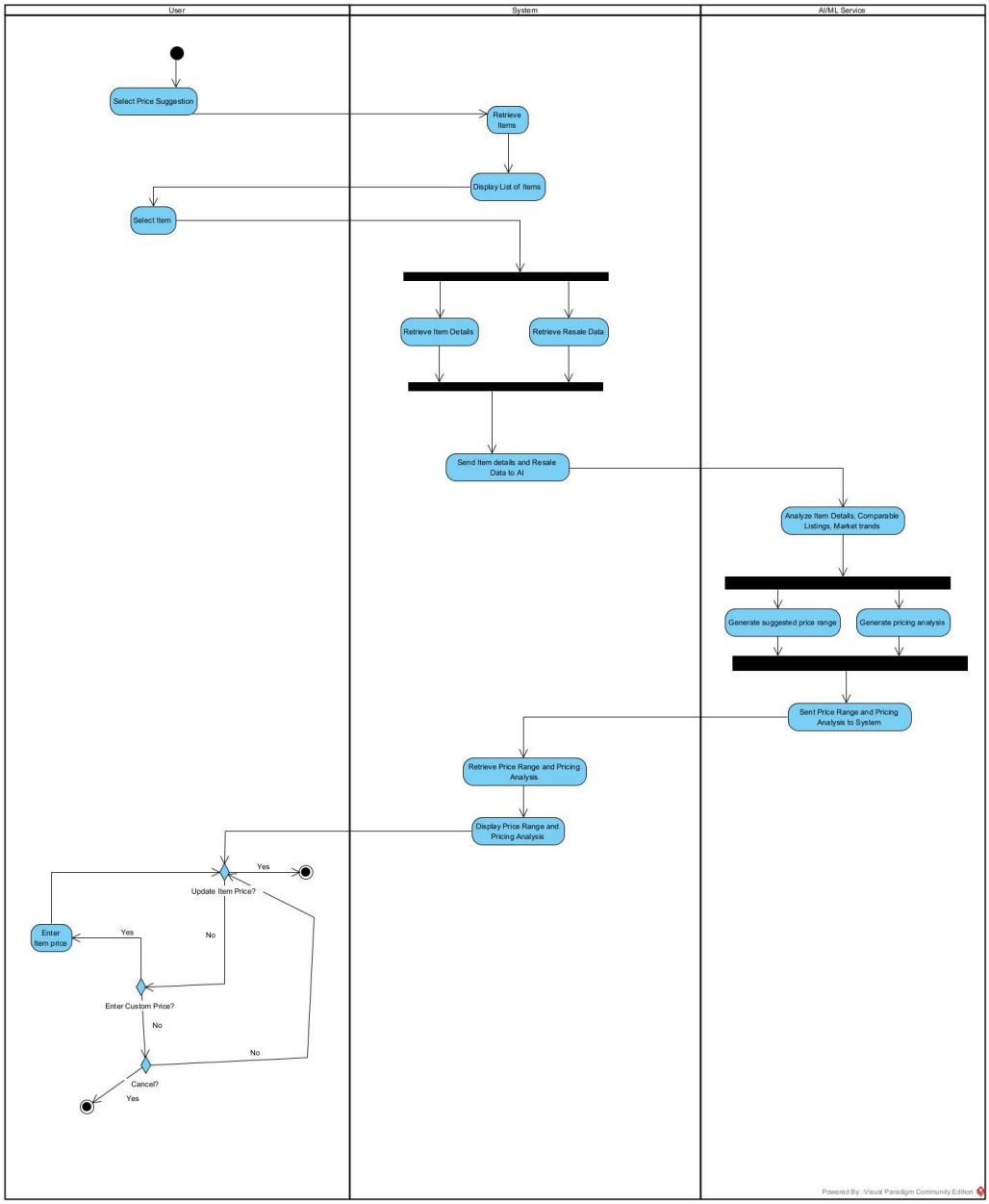
AD2:


User
System
Select Add Item
Retrieve Item Categories
Display list of Item Categories
Select Item Category
Enter Brand, condition, and description
Upload Item image
No
No
No
Save?
Yes
Valid?
Yes
Cancel?
Yes
Send Item Details to AI model
Display Item Details
Powered By Visual Paradigm Community Edition

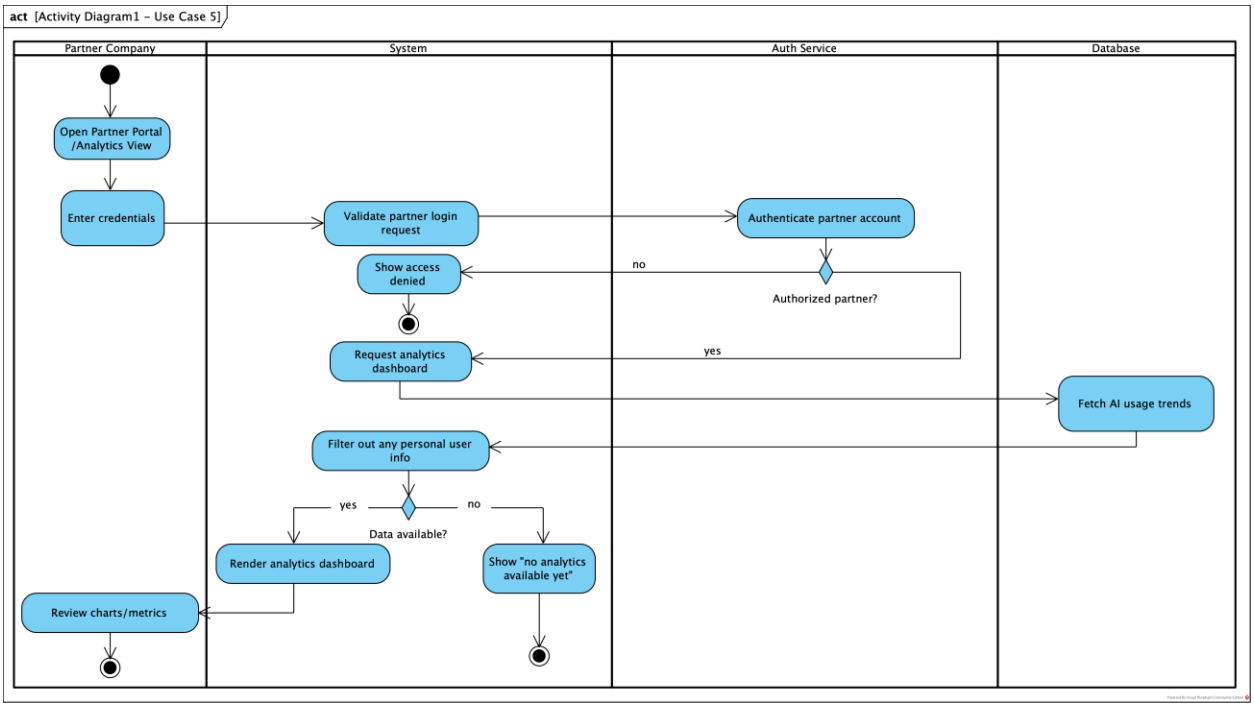
AD3:


act [Activity Diagram1 – Use Case 3]
User
System
Partner Platforms (Poshmark/Depop/Karrot)
User
Open Saved Item
Type "Market Comparison"
Prepare query using item attributes
Fetch comparable listings/prices
Aggregate results
Compute Average/Range
Derive insight (above/below/within market)
no
yes
Partner data available?
Show partial results + warning
Show full comparison
View market trends and comparable prices

AD4:


User
System
AI/ML Service
Select Price Suggestion
Retrieve Items
Display List of Items
Select Item
Retrieve Item Details
Retrieve Resale Data
Send Item details and Resale Data to AI
Analyze Item Details, Comparable Listings, Market trands
Generate suggested price range
Generate pricing analysis
Sent Price Range and Pricing Analysis to System
Retrieve Price Range and Pricing Analysis
Display Price Range and Pricing Analysis
Yes
Update Item Price?
Enter Item price
Yes
No
Enter Custom Price?
No
No
Cancel?
Yes
Powered By Visual Paradigm Community Edition

AD5:

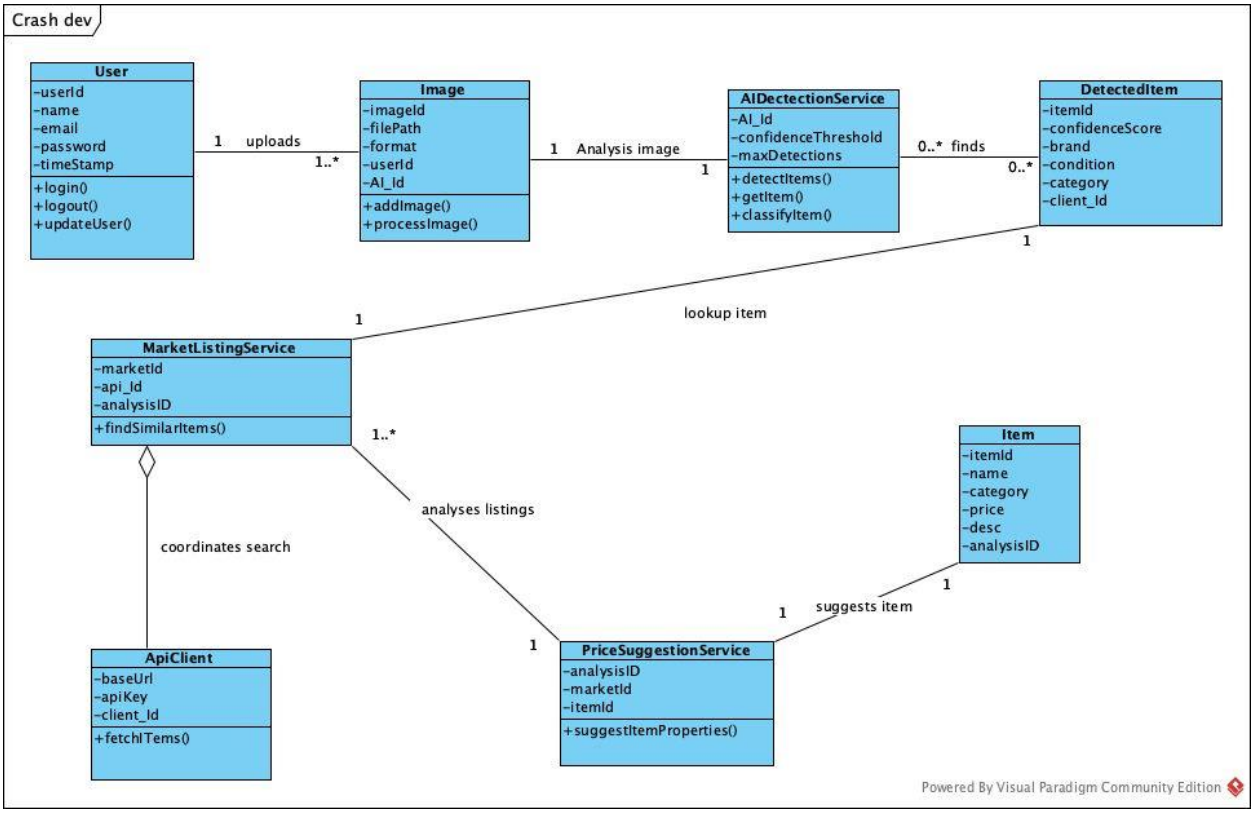


3.4 Business Rules

Business Rule #	Description	Activity Diagram	Related UCS	UI Mock-up
BR1	The system must authenticate users using valid email and password credentials before granting access to protected features.	AD1	UC1	UI 2.7.2
BR2	User must enter category, brand, condition, and at least one image before any analysis can be done.	AD2	UC2	UI 2.7.3
BR3	The system will use partner resale data for pricing analysis and will not store or expose personal information from user.	AD3	UC3	UI 2.7.4
BR4	Market price, price range, and comparable listings must be shown when displaying market insights.	AD3	UC3	UI 2.7.4
BR5	AI/ML will generate price suggestions based only on the given item details, historical sales data, and real-time market data.	AD4	UC4	UI 2.7.6

BR6	Users may modify or reject AI-generated price suggestions before saving the final listing price.	AD4	UC4	UI 2.7.6
BR7		AD7	UC6	UI 2.7.7
BR8		AD8	UC7	UI 2.7.8
BR9		AD8	UC7	UI 2.7.8
BR10		AD8	UC7	UI 2.7.8
BR11		AD8	UC7	UI 2.7.8
BR12		AD8	UC7	UI 2.7.8
BR13		AD9	UC8	UI 2.7.9
BR14		AD9	UC8	UI 2.7.9
BR15		AD9	UC8	UI 2.7.9
BR16		AD9	UC8	UI 2.7.9
BR17		AD10	AD9	UI 2.7.9
BR18		AD10	AD9	UI 2.7.9
BR19		AD10	AD9	UI 2.7.9

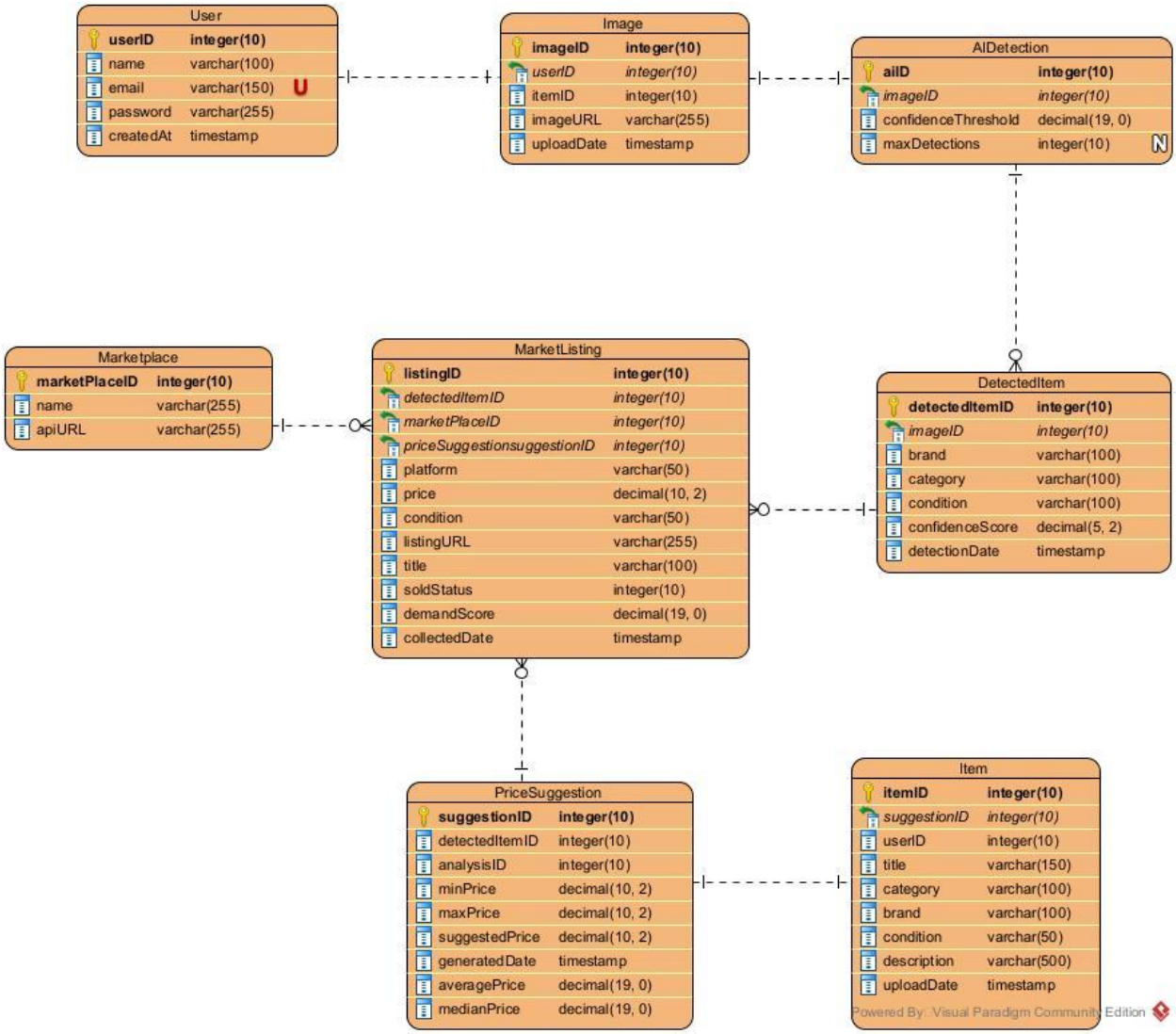
Section 4 – Domain Class



Section 5 – Database

The FlipValue system uses a relational database to store user information, item details, uploaded images, AI analysis results, price suggestions, and comparable market listings. The database schema consists of six main entities: User, Item, Image, AIAnalysis, PriceSuggestion, and MarketListing.

Primary keys uniquely identify each record, while foreign keys establish relationships between tables to maintain referential integrity. This structure supports the core functionality of the application, including item management, AI-assisted pricing analysis, and cross-platform market comparison.



Data Dictionary

User

Attribute	Data type	Key	Description
userID	INT	PK	Unique identifier for each user
name	VARCHAR(100)		User's full name
email	VARCHAR(100)	UNIQUE	User login email
password	VARCHAR(255)		Encrypted Password
createdAt	TIMESTAMP		Date account was created

Image

Attribute	Data type	Key	Description

imageID	INT	PK	Unique image identifier
userID	INT	FK	User who uploaded image
itemID	INT	FK	Item associated with image
imageURL	VARCHAR(255)		Location of image
uploadDate	TIMESTAMP		Date image was uploaded

AIDetection

Attribute	Data type	Key	Description
aiID	INT	PK	AI Image detection ID
imageID	INT	FK	Image analyzed
confidenceThreshold	DECIMAL		Minimum confidence for detection
maxDetections	INT		Maximum number of detected items

DetectedItem

Attribute	Data type	Key	Description
detectedItemID	INT	PK	Unique detected item identifier
imageID	INT	FK	Image with detected item
brand	VARCHAR(100)		Detected brand
category	VARCHAR(100)		Detected category
condition	VARCHAR(100)		Estimated item condition
confidenceScore	DECIMAL(5,2)		AI confidence score
detectionDate	TIMESTAMP		Detection timestamp

PriceSuggestion

Attribute	Data type	Key	Description
suggestionID	INT	PK	Unique price suggestion identifier
detectedItemID	INT	FK	Detected item reference
analysisID	INT	FK	AI analysis reference
minPrice	DECIMAL(10,2)		Min market price
maxPrice	DECIMAL(10,2)		Max market price

suggestedPrice	DECIMAL(10,2)		Recommended selling price
generatedDate	TIMESTAMP		Date suggestion was generated
averagePrice	DECIMAL(10,2)		Average market price
medianPrice	DECIMAL(10,2)		Median market price

Marketplace

Attribute	Data type	Key	Description
marketPlaceID	INT	PK	Unique marketplace identifier
name	VARCHAR(255)		Marketplace name
apiURL	VARCHAR(255)		API endpoint

MarketListing

Attribute	Data type	Key	Description
listingID	INT	PK	Unique listing identifier
detectedItemID	INT	FK	Detected item
marketPlaceID	INT	FK	Marketplace where listing appears
priceSuggestionID	INT	FK	Price suggestion
platform	VARCHAR(50)		Listing platform
price	DECIMAL(10,2)		Listing price
condition	VARCHAR(50)		Item condition
listingURL	VARCHAR(255)		Link to listing
title	VARCHAR(100)		Listing title
soldStatus	INT		Whether item is sold
demandScore	DECIMAL		Estimated demand score
collectedDate	TIMESTAMP		Date listing data collected

Item

Attribute	Data type	Key	Description
itemID	INT	PK	Unique item identifier
priceSuggestionID	INT	FK	Reference to price suggestion
userID	INT	FK	User who uploaded item
title	VARCHAR(150)		Item title

category	VARCHAR(100)	Item category
brand	VARCHAR(100)	Item brand
condition	VARCHAR(50)	Condition of item
description	VARCHAR(500)	Item description
uploadDate	TIMESTAMP	Date item uploaded to marketplace

Section 6 – Project Management

6.1 Work Breakdown Structure

Level 1: Product Goal (Epic Level)

1. **FlipValue: Resale Pricing App**

Level 2: Major Deliverables (High-Level Features / Epics)
1.1 **User Account Management**
1.2 **Item Upload and Management**
1.3 **AI Price Suggestion System**
1.4 **Market Data Integration**
1.5 **Dashboard and Insights**
1.6 **External Marketplace Redirection**
1.7 **UI/UX Design**
1.8 **Backend and Database**
1.9 **Testing and Quality Assurance**
1.10 **Deployment**

Level 3: Iteration/Sprint-Level Breakdown (User Stories/Features)

- **1.1 User Account Management**
  - 1.1.1 As a user, I can register for an account
  - 1.1.2 As a user, I can log into my account
  - 1.1.3 As a user, I can log out of my account
  - 1.1.4 As a user, I can view my profile information
  - 1.1.5 As a user, I can edit m profile information
  - 1.1.6 As a user, I can reset my password if I forget
  - 1.1.7 As a user, I can delete my account
- **1.2 Item Upload and Management Module**

  - 1.2.1 As a user, I can upload images of my item using my device camera or photo gallery
  - 1.2.2 As a user, I can save item listings for future reference
  - 1.2.3 As a user, I can edit item details after saving
  - 1.2.4 As a user, I can delete an item listing
  - 1.2.5 As a user, I can view a list of all my saved items
  - 1.2.6 As a user, I can filter my items based on their condition, brand, and price ranges
  - 1.2.7 As a user, I can check off an item as either available, sold, or pending
- **1.3 AI Price Suggestion System**
  - 1.3.1 As a user, I can submit a picture of my item for AI price analysis
  - 1.3.2 As a user, I can receive suggested price range for my item
  - 1.3.3 As a user, I can view comparable market listings used for the price suggestion
  - 1.3.4 As a user, I can see pricing trends (average price, demand, etc.)
  - 1.3.5 As a user, I can see an analysis of how the suggested price was calculated
  - 1.3.6 As a user, I can update item details and request a new price suggestion and edit it
- **1.4 Market Data Integration**
  - 1.4.1 As a user, I can connect to external APIs
  - 1.4.2 As a user, I can view average selling prices or similar items
  - 1.4.3 As a user, I can see demand trends for my item category
  - 1.4.4 As a user, I can see past price data for similar items
- **1.5 Dashboard and Insights**
  - 1.5.1 As a user, I can view all my items in one dashboard
  - 1.5.2 As a user, I can see the suggested prices for my items
  - 1.5.3 As a user, I can view my market trends and analytics
  - 1.5.4 As a user, I can filter and sort my items
  - 1.5.5 As a user, I can view my item performance insights
- **1.6 External Marketplace Redirection**
  - 1.6.1 As a user, I can export my item details for marketplace listing
  - 1.6.2 As a user, I can be redirected to external platforms to complete my listing
- **1.7 UI/UX Design**
  - 1.7.1 As a user, I can navigate the app easily using a clear interface
  - 1.7.2 As a user, I can view item information in a clear layout
  - 1.7.3 As a user, I can use the app on mobile devices
  - 1.7.4 As a developer, I can create mock-ups for the app
- **1.8 Backend and Database**
  - 1.8.1 As a system, I can store user account information securely
  - 1.8.2 As a system, I can store item details and images
  - 1.8.3 As a system, I can store AI pricing results
  - 1.8.4, As a system, I can retrieve item and pricing data when requested
- **1.9 Testing and Quality Assurance**

    - 1.9.1 As a developer, I can test individual system components
    - 1.9.2 As a developer, I can test system integration
    - 1.9.3 As a user, I can test the system and provide feedback
- **1.10 Deployment**
    - 1.10.1 As a developer, I can track the deployment's version history so that I can improve the app's performance
    - 1.10.2 As a user, I can trust that my data will be secured safely after each deployment, so that my personal info is protected
    - 1.10.3 As a user, I can successfully access the mobile application, so that I can start pricing my items
    - 1.10.4 As a developer, I can be informed of a failed deployment, so that I can figure out what went wrong and troubleshoot effectively

Level 4: Sprint Tasks (Granular Task-Level WBS)
For **1.2.1 Enter Item Details**, we break into actual tasks:

- **1.1.1 User registration**
    - 1.1.1.1 Design registration UI (email, username, password)
    - 1.1.1.2 Implement fronted form validation (required fields)
    - 1.1.1.3 Create backend API for user registration
    - 1.1.1.4 Store user data in database
    - 1.1.1.5 Implement password encryption
    - 1.1.1.6 Test user registration functionality
- **1.1.2 User Login**
    - 1.1.2.1 Design login screen UI
    - 1.1.2.2 Implement login form validation
    - 1.1.2.3 Create backend login information
    - 1.1.2.4 Verify user credentials from database
    - 1.1.2.5 Implement session management
    - 1.1.4.6 Test login functionality
- **1.1.3 Logout**
    - 1.1.3.1 Add logout button to UI
    - 1.1.3.2 Implement logout function
    - 1.1.3.3 Destroy user session
    - 1.1.3.4 Redirect user to login page
    - 1.1.3.5 Test logout functionality
- **1.1.4 View Profile**
    - 1.1.4.1 Design profile page UI
    - 1.1.4.2 Retrieve user data from database
    - 1.1.4.3 Display user information on profile page

  - 1.1.4.4 Test profile display
- **1.1.5 Edit Profile**
  - 1.1.5.1 Design edit profile page
  - 1.1.5.2 Implement email verification
  - 1.1.5.3 Create backend API to update profile
  - 1.1.5.4 Update user data in database
  - 1.1.5.5 Display update confirmation message
  - 1.1.5.6 Test edit profile functionality

- **1.2.1 Upload Item Images**
  - 1.2.1.1 Design image upload UI (camera, photo gallery)
  - 1.2.1.2 Implement image selection from device
  - 1.2.1.3 Implement image upload to server/cloud
  - 1.2.1.4 Link uploaded images to item record in database
  - 1.2.1.5 Validate file type and size
  - 1.2.1.6 Test image upload functionality
- **1.2.2 Save Item Listing**
  - 1.2.2.1 Implement save button functionality
  - 1.2.2.2 Ensure item data is stored in database
  - 1.2.2.3 Confirm successful save (pop up message)
  - 1.2.2.4 Handle save errors (missing data)
  - 1.2.2.5 Test saving functionality
- **1.2.3 Edit Item**
  - 1.2.3.1 Design edit item UI
  - 1.2.3.2 Load existing item data into form
  - 1.2.3.3 Update backend API for editing items
  - 1.2.3.4 Save updated data to database
  - 1.2.3.5 Test edit functionality
- **1.2.4 Delete Item**
  - 1.2.4.1 Add delete button to item view
  - 1.2.4.2 Implement confirmation prompt
  - 1.2.4.3 Create backend delete API
  - 1.2.4.4 Remove Item from database
  - 1.2.4.5 Tests delete functionality
- **1.2.5 View Saved Items**
  - 1.2.5.1 Design item list UI (dashboard/list)
  - 1.2.5.2 Fetch items from database
  - 1.2.5.3 Display item details (image, name, price suggestion)
  - 1.2.5.4 implement filtering and sorting
  - 1.2.5.5 Test item display functionality
- **1.3.1 Submit Item for AI Analysis**

- 1.3.1.1 Capture item details and images from user
- 1.3.1.2 Validate input (required fields, image format)
- 1.3.1.3 Send item data to backend API
- 1.3.1.4 Store item data temporarily for processing
- 1.3.1.5 Trigger AI analysis process
- 1.3.1.6 Test data submission functionality

- **1.3.2 Generate AI Price Suggestion**
  - 1.3.2.1 Retrieve item attributes (brand, condition, category)
  - 1.3.2.2 Fetch comparable item listings from external APIs
  - 1.3.2.3 Process past sales data
  - 1.3.2.4 Apply pricing algorithm / AI model
  - 1.3.2.5 Calculate suggested price range
  - 1.3.2.6 Store generated price results in database
  - 1.3.2.7 Test pricing accuracy and functionality
- **1.3.3 View Comparable Listings**
  - 1.3.3.1 Retrieve comparable item data from backend
  - 1.3.3.2 Format listing data (price, condition, platform)
  - 1.3.3.3 Display comparable listings in UI
  - 1.3.3.4 Test data display functionality
- **1.3.4 Display Pricing Trends**
  - 1.3.4.1 Retrieve price data history from external APIs
  - 1.3.4.2 Calculate average price as well as demand
  - 1.3.4.3 Display trends on the dashboard
  - 1.3.4.4 Test pricing trend display functionality
- **1.3.5 View Comparable Listings**
  - 1.3.5.1 Identify factors affecting price (condition, demand, brand)
  - 1.3.5.2 Generate explanation text
  - 1.3.5.3 Display explanation in UI
  - 1.3.5.4 Test explanation clarity and accuracy
- **1.3.6 Price Suggestion with User Edits**
  - 1.3.6.1 Allow user to update item details
  - 1.3.6.2 Resubmit updated data to AI system
  - 1.3.6.3 Recalculate price suggestion
  - 1.3.6.4 Update stored pricing results
  - 1.3.6.5 Test re-analysis functionality
- **1.4.1 Connect to Partner APIs**

  - 1.4.1.1 Identify partner platforms and available APIs
  - 1.4.1.2 Review API documentation and access requirements
  - 1.4.1.3 Configure API keys and authentication
  - 1.4.1.4 Implement backend connection to partner APIs
  - 1.4.1.5 Test API connection functionality
- **1.4.2 View Average Selling Prices/Items**
  - 1.4.2.1 Retrieve average price data from external APIs
  - 1.4.2.2 Process and calculate the averages from the retrieved data
  - 1.4.2.3 Display the average prices
  - 1.4.2.4 Test the accuracy and display of the average prices
- **1.4.3 View Demand Trends**
  - 1.4.3.1 Retrieve listing frequency data from APIs
  - 1.4.3.2 Analyze number of similar items in market
  - 1.4.3.3 Evaluate demand indicators
  - 1.4.3.4 Generate trends summaries
  - 1.4.3.5 Display demand trends in dashboard
  - 1.4.3.5 Test demand analysis functionality
- **1.4.4 View Past Price Data**
  - 1.4.4.1 Retrieve past price data from available APIs
  - 1.4.4.2 Display past prices
  - 1.4.4.3 Test and demo past price display data
- **1.5.1 View All Items**
  - 1.5.1.1 Design and create the dashboard
  - 1.5.1.2 Get all user items from the database
  - 1.5.1.2 Show all item details such as images, name, price, etc.
  - 1.5.1.3 Add sorting and filtering functionality
  - 1.5.1.4 Test dashboard functionality
- **1.5.2 Display Suggested Price**
  - 1.5.2.1 Get AI price suggestions per item
  - 1.5.2.2 Show pricing suggestions beside corresponding item
  - 1.5.2.3 Test and validate pricing suggestions

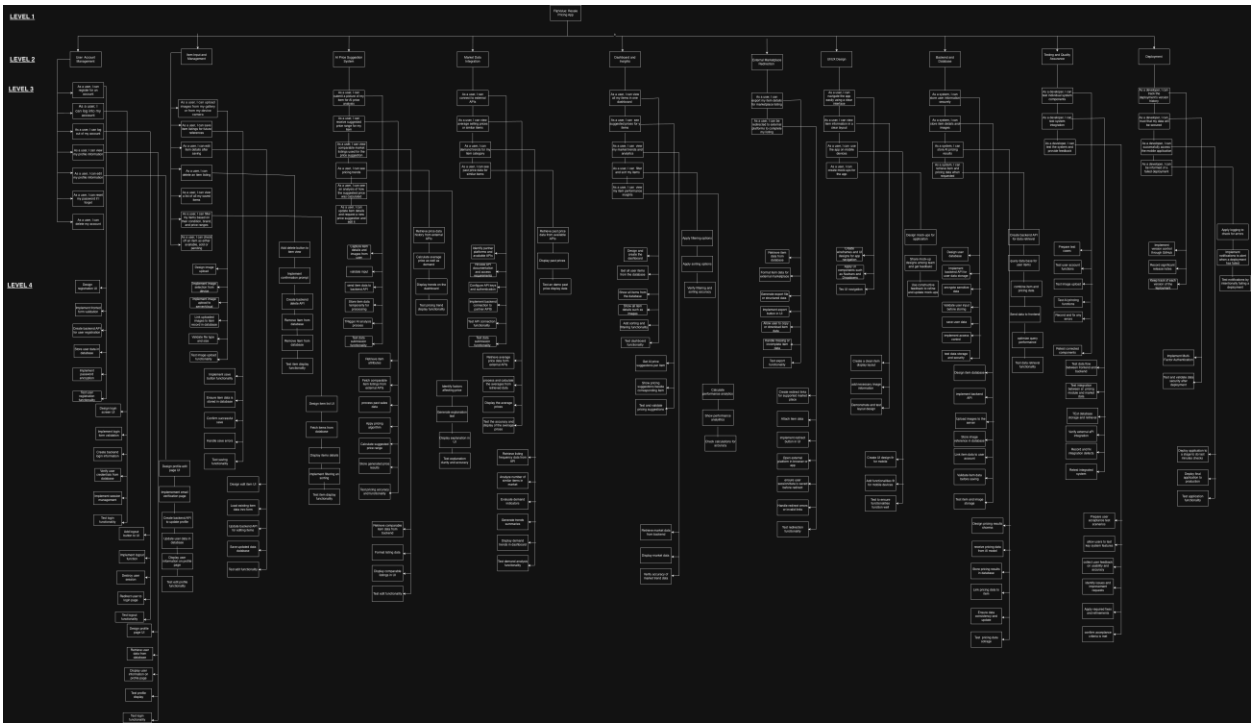
- **1.5.3 Display Market Analytics**
  - 1.5.3.1 Retrieve market data from backend
  - 1.5.3.2 Display market data
  - 1.5.3.3 Verify accuracy of market trend data
- **1.5.4 Filter Items**
  - 1.5.4.1 Apply filtering options such as price, category, brand, etc
  - 1.5.4.2 Apply sorting options such as alphabetical order, price, etc
  - 1.5.4.3 Verify filtering and sorting accuracy
- 1.5.5 Display Item Performance Insights
  - 1.5.5.1 Calculate performance analytics such as views, or any price changes
  - 1.5.5.2 Show performance analytics
  - 1.5.5.3 Check calculations for accuracy
- **1.6.1 Export Item Details**
  - 1.6.1.1Retrieve item data from database (title, description, price, images)
  - 1.6.1.2 Format item data for external marketplace requirements
  - 1.6.1.3 Generate export file or structured data
  - 1.6.1.4 Implement export button in UI
  - 1.6.1.5 Allow user to copy or download item details
  - 1.6.1.6 Handle missing or incomplete item data
  - 1.6.1.7 Test export functionality
- **1.6.2 Redirect to External Platforms**
  - 1.6.2.1 Create redirect links for supported marketplace (eBay, Depop, Karrot, etc.)
  - 1.6.2.2 Attach item data (if supported by platform URL structure)
  - 1.6.2.3 Implement redirect button in UI
  - 1.6.2.4 Open external platform in browser or app
  - 1.6.2.5 Ensure user session/data is saved before redirect
  - 1.6.2.6 Handle redirect errors or invalid links
  - 1.6.2.7 Test redirection functionality
- 1.7.1 User-Friendly App Navigation
  - 1.7.1.1 Create wireframes and UI designs for app navigation
  - 1.7.1.2 Apply UI components such as Navbars and Dropdowns

    - 1.7.1.3 Test UI navigation components and wireframes
- 1.7.2 Structured Item Info Layout
    - 1.7.2.1 Create a clean item display layout
    - 1.7.2.2 Add necessary image info such as image, name, and price
    - 1.7.2.3 Demo and test layout design
- 1.7.3 Mobile Compatibility
    - 1.7.3.1 Create UI design fit for mobile
    - 1.7.3.2 Add functionalities that aren't available on websites, such as touch controls and push notifications
    - 1.7.3.3 Test to ensure that the app is accessible on mobile and everything fits to scale
- 1.7.4 Create Mock-ups
    - 1.7.4.1 Design mock-ups for application
    - 1.7.4.2 Share mock-up designs among team and receive feedback
    - 1.7.4.3 Use constructive feedback to refine and update mock-ups
- **1.8.1 Store User Account Information Securely**
    - 1.8.1.1 Design user database
    - 1.8.1.2 Implement backend API for user data storage
    - 1.8.1.3 Encrypt sensitive data (password)
    - 1.8.1.4 Validate user input before storing
    - 1.8.1.5 Save user data to database
    - 1.8.1.6 Implement access control and authentication
    - 1.8.1.7 Test data storage and security
- **1.8.2 Store Item Details and Images**
    - 1.8.2.1 Design item database schema (category, brand, condition, etc.)
    - 1.8.2.2 Implement backend API for item submission
    - 1.8.2.3 Upload images to server
    - 1.8.2.4 Store image references/URLs in database
    - 1.8.2.5 Link item data to user account
    - 1.8.2.6 Validate item data before saving
    - 1.8.2.7 Test item and image storage

- **1.8.3 Store AI Pricing Results**
  - 1.8.3.1 Design pricing results schema (price range, trends, factors)
  - 1.8.3.2 Receive pricing data from AI model
  - 1.8.3.3 Store pricing results in database
  - 1.8.3.4 Link pricing data to corresponding item
  - 1.8.3.5 Ensure data consistency and updates
  - 1.8.3.6 Test pricing data storage
- **1.8.4 Retrieve Item and Pricing Data**
  - 1.8.4.1 Create backend API for data retrieval
  - 1.8.4.2 Query database for user items and pricing results
  - 1.8.4.3 Combine item and pricing data
  - 1.8.4.4 Send data to frontend for display
  - 1.8.4.5 Optimize query performance
  - 1.8.4.6 Test data retrieval functionality
- **1.9.1 Testing Individual System Components**
  - 1.9.1.1 Prepare test case for each module
  - 1.9.1.2 Test user account functions
  - 1.9.1.3 Test item input and image upload functions
  - 1.9.1.4 Test AI pricing functions
  - 1.9.1.5 Record and fix any errors or bugs
  - 1.9.1.6 Retest corrected components
- **1.9.2 Test System Integration**
  - 1.9.2.1 Test data flow between frontend and backend
  - 1.9.2.2 Test integration between AI pricing module and market data
  - 1.9.2.3 Test database storage and retrieval
  - 1.9.2.4 Verify external API integration
  - 1.9.2.5 Record and fix integration defects
  - 1.9.2.6 Retest integrated system
- 1.9.3 Test System and Collect User Feedback
  - 1.9.3.1 Prepare user acceptance test scenarios
  - 1.9.3.2 Allow users to test key system features

- 1.9.3.3 Collect user feedback on usability and accuracy
- 1.9.3.4 Identify issues and improvement requests
- 1.9.3.5 Apply required fixes and refinements
- 1.9.3.6 Confirm acceptance criteria is met

- **1.10.1 Track Deployment History**
  - 1.10.1.1 Implement version control through GitHub
  - 1.10.1.2 Record significant release notes
  - 1.10.1.3 Keep track of each version of the deployment
- **1.10.2 Data Security**
  - 1.10.2.1 Implement Multi-Factor Authentication to prevent unauthorized personnel form accessing info
  - 1.10.2.2 Test and validate data security after deployment
- **1.10.3 Application Accessibility**
  - 1.10.3.1 Deploy application to a stage to do last minute checks
  - 1.10.3.2 Deploy final application to production, to allow for user access
  - 1.10.3.3 Test application functionality
- **1.10.4 Failed Deployment**
  - 1.10.4.1 Apply logging to check for errors
  - 1.10.4.2 Implement notifications to alert when a deployment has failed
  - 1.10.4.3 Test notifications by intentionally failing a deployment



6.2 Milestones & Acceptance Criteria

Project Milestones & Acceptance Criteria

Milestone	Timeframe	WBS Mapping	Purpose	Acceptance Criteria
1. Project Kickoff & Design Alignment	Week 2	1.7 UI/UX Design, 1.1 User Account Management	Align team, finalize design, prepare backlog	UI/UX mockups approved; backlog created; Agile meetings scheduled; dev environment set up
2. Core User Account System	Week 4	1.1 User Account Management, 1.8 Backend & Database	Implement authentication & user profiles	Users can register/login/logout; data stored securely; profile updates work; demo approved
3. Item Input & Management	Week 6	1.2 Item Input and Management	Enable item creation and management	Enter details; upload images; save/edit/delete works; items display; features tested
4. AI Price Suggestion System	Week 8	1.3 AI Price Suggestion System, 1.4 Market Data Integration	Implement AI-based pricing	AI analyzes items; comparable listings retrieved; price range generated; users can adjust
5. Dashboard & Market Insights	Week 10	1.5 Dashboard and Insights	Provide insights and item overview	Dashboard shows items; filtering/sorting works; comparisons visible; pricing clarity
6. MVP Integration & Release	Week 12	1.6 External Marketplace Redirection, 1.10 Deployment	Deliver functional MVP	Core features integrated; marketplace links work; deployed to test env; demo + feedback
7. Testing & QA	Week 14	1.9 Testing and Quality Assurance	Ensure system quality and reliability	All tests pass; no critical bugs; performance acceptable; security validated

8. Final Release & Closure	Week 16	1.10 Deployment	Deliver final product and close project	Deployed to production; features complete; final demo approved; documentation done

Section 7 – Product Backlog & Implementation Schedule

Section 8 – Client/Faculty Sign-off